Yunxin Sun

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Research Interests

LLM agents, evaluation methods, post-training methods, and their applications in science.

Education Background

Ph.D. in Computer Science — West Lafayette, IN
Purdue University — *Expected graduate: May 2028*

M.Sc. in Computer Science — Zurich, Switzerland
ETH Zurich

B.Eng. in Software Engineering — Shanghai, China
Tongji University

Experience

External Research Collaborator — Remote
Nace.ai
I collaborated with Nace.ai on developing a hypernetwork-based approach for test-time adaptation. I first curated a large-scale training dataset comprising 4 million counterfactual reasoning questions with long-context inputs. I then trained a hypernetwork in a distributed setting to dynamically generate a context-specific LoRA adapter for each input. Our approach outperformed several model editing and in-context learning baselines while achieving lower inference costs.

Research Internship — San Jose, CA
TikTok US Infrastructure Lab
During my internship at TikTok, I investigated performance bottlenecks in RDMA-based distributed memory transfers within internal database systems. I collaborated closely with cross-functional team members to diagnose slow queries and identify underlying performance issues. My investigation spanned multiple system layers, including software architecture, network protocols, and hardware-level primitives. I extensively leveraged performance profiling and observability tools, such as eBPF, to analyze system behavior and pinpoint bottlenecks. I also contributed to developing and optimizing other database functionality throughout the internship.

Publications

- **Yunxin Sun**, Abulhair Saparov. "$(\mathcal{LC})^2$: *L*earning *C*onstant-size *L*atent *C*ontext for Language Model Agents ". *preprint*.
- **Yunxin Sun**, Abulhair Saparov. "Do Language Models Follow Occam's Razor? An Evaluation of Parsimony in Inductive and Abductive Reasoning". In EMNLP 2026.
- Revanth Rameshkumar, Jimson Huang, **Yunxin Sun**, Fei Xia, Abulhair Saparov. "Reasoning Models Reason Well, Until They Don't". In AACL-IJCNLP 2025
- Jinghan Sun, Shaobo Li, **Yunxin Sun**, Chao Sun, Dejan Vucinic, and Jian Huang. "LeaFTL: A Learning-Based Flash Translation Layer for Solid-State Drives." In ASPLOS 2023.
- **Yunxin Sun**, Tamar Shinar, and Craig Schroeder. "Effective time step restrictions for explicit MPM simulation." In Computer Graphics Forum, vol. 39, no. 8, pp. 55-67. 2020.
- Josh Urban Davis, Jun Gong, **Yunxin Sun**, Parmit Chilana, and Xing-Dong Yang. "Circuitstyle: A system for peripherally reinforcing best practices in hardware computing." In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology, pp. 109-120. 2019.
- Guokai Zhang, Ye Luo, Dandan Zhu, Yixuan Xu, **Yunxin Sun**, and Jianwei Lu. "Spatial pyramid dilated network for pulmonary nodule malignancy classification." In 2018 24th International Conference on Pattern Recognition (ICPR), pp. 3911-3916. IEEE, 2018.

Other Manuscripts

- "Execution Time Prediction For Serverless Workload". Technical Report.

Teaching and Mentoring Experience

- Gave lectures about *LLM Agents* and *Reinforcement Learning*, in the **graduate** course "*Natural Language Processing*" at Purdue University.
- Courses I TAed at Purdue: Natural Language Processing (**graduate**), Large Scale Data Analytics, Data Engineering I & II, Programming With Multimedia Objects, C Programming, Information Systems.
- Mentored two undergraduate students in the course "*Seminar on Modern SSDs*" at ETH Zurich. Met with them weekly to check the progress of the research project.
- Gave a lecture about *MQSim*, a state-of-the-art simulator for Solid State Drives, in the course "*Seminar on Modern SSDs*" at ETH Zurich.

Skills

- **Programming Languages**: Python, C, C++, SQL, Java, Bash.
- **Tools & Platforms**: PyTorch, Hugging Face (Transformers, TRL, PEFT, Accelerate), vLLM, DeepSpeed, NumPy, scikit-learn, Git, Docker, Linux, SLURM